METRIC
A-A-59398
09 January 2004

COMMERCIAL ITEM DESCRIPTION

# GASKET MATERIAL, SILICONE GEL WITH MONEL MATRIX OR EQUIVALENT

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description covers the requirements for silicone gel gasket material with a Monel® matrix or equivalent, used to provide environmental protection, electromagnetic shielding, and electrical bonding for joints between surfaces for electrical and electronic installations in shipboard and other exposed applications. The gaskets are applied to joints for which the surfaces have been prepared by cleaning to base metal.

2. CLASSIFICATION. The gaskets shall be of the following types, classes, and sizes:

2.1 Types and classes.

Type I - Sheet gasket material (field-cut to size).

Class 1 - Standard silicone gel gasket.

Class 2 - High-performance silicone gel gasket, high salt spray resistance.

Class 3 - Fluorosilicone gel gasket, high fuel and solvent resistance.

Type II - Strip or cord gasket (field-cut to length), high-performance silicone gel, Monel® or equivalent knitted rope gasket.

Type III - Pre-cut or pre-formed gaskets.

2.2 Sizes. Gasket material shall be shall be specified by citation of the following dimensions for each type. The currently available sizes are shown in section 7.1.

Type I - XX mm (xx in) length by YY mm (yy in) width by ZZ mm (zz in) thick.

Type II - XX mm (xx in) diameter by YY mm (yy in) spool length.

Type III - Custom cut per user's requirements.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05Q, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to commandstandards@navsea.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

FSC 2090

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3. SALIENT CHARACTERISTICS.

3.1 Materials. The sleeves shall be fabricated from a Monel® matrix or equivalent, impregnated with a soft, cross-linked silicone gel. The gel provides a water-repelling environmental seal; the Monel® matrix or equivalent provides bonding and shielding. The use of the Monel® matrix or equivalent is required to provide protection against galvanic corrosion in the marine and exposed environments. Any change in the matrix material shall be considered as non-conformance with the salient characteristics of this commercial item description.

3.2 Operating environment. The gaskets are intended for use with equipment installations in exposed, unsheltered environments aboard surface ships and in exposed environments in shore and land-mobile applications. The sleeves shall remain fully functional for the temperature range -25 to 125 °C. The gasket material shall remain fully functional when gasketed joints are exposed to solar radiation, salt spray, and stack/exhaust gases. The design service life for the specified operating environment shall be for the service life of the joint, but not less than 20 years.

3.3 Performance. The gaskets shall have the following performance characteristics:

a. Seal, waterproof, and provide bonding for bi-metallic joints (see 7.4.b).

b. Tack instantly to prepared surfaces (see 7.4.b).

c. Be removable for re-entry or opening of the joint (see 7.4.b).

d. Be installed without personal protective equipment or special tools (see 7.4.b).

e. Provide protection against galvanic corrosion (see 7.4.b).

f. Shield against electromagnetic interference (EMI) (see 7.4.b).

g. Provide low electrical resistance across the joint (see 7.4.b).

h. Require no cure cycle (see 7.4.b).

3.4 Assembly. Type I and II gasket material shall be provided in bulk. Type III (pre-cut or pre-formed) gaskets shall be provided in bulk and packed so as to provide protection for gaskets remaining in the package. Each package shall include the gasket material and an instruction sheet.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE.

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

6. PACKAGING.

6.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or order.

6.2 Shelf life. The shelf life of the gasket material shall be not less than 60 months when stored in a sheltered space having a controlled environment.

6.3 Labeling. The kits shall be labeled as to dimensions.



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7. NOTES.

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.

This example PIN code is for a type I high-performance silicone gel gasket, high salt spray resistance, with thickness 1.1 mm, length 175 m, and width 200 mm: AA59398-A2H.

PIN code:

| **AA59398** | – | **X** | **X** | **XX** |
|---|---|---|---|---|
| CID number | | Gasket type (see code below) | Gasket class (see code below) | Gasket size (see code below) |

| **Product** | **Thickness [mm (in)]** | **Length [mm (in)]** | **Width [mm (in)]** | **National Stock Number** | **Gasket Type PIN Code** | **Gasket Class PIN Code** | **Gasket Size PIN Code** |
|---|---|---|---|---|---|---|---|
| Type I, Class 1 | 0.66 (0.026) | 175 (7.0) | 200 (8.0) | TBD | A | 1 | A |
| | 0.66 (0.026) | 175 (7.0) | 375 (15.0) | TBD | A | 1 | B |
| | 0.66 (0.026) | 350 (14.0) | 500 (20.0) | TBD | A | 1 | C |
| | 0.66 (0.026) | 600 (24.0) | 600 (24.0) | TBD | A | 1 | D |
| | 1.10 (0.044) | 175 (7.0) | 200 (8.0) | TBD | A | 1 | E |
| | 1.10 (0.044) | 800 (32.0) | 190 (7.5) | TBD | A | 1 | F |
| | 1.10 (0.044) | 2400(96.0) | 600 (24.0) | TBD | A | 1 | G |
| Type I, Class 2 | 1.10 (0.044) | 175 (7.0) | 200 (8.0) | TBD | A | 2 | H |
| | 1.10 (0.044) | 175 (7.0) | 375 (15.0) | TBD | A | 2 | I |
| | 1.10 (0.044) | 575 (23.0) | 575 (23.0) | TBD | A | 2 | J |
| Type I, Class 3 | 1.10 (0.044) | 175 (7.0) | 200 (8.0) | TBD | A | 3 | K |
| | 1.10 (0.044) | 175 (7.0) | 375 (15.0) | TBD | A | 3 | L |
| Type II | **Thickness [mm (in)]** | **Length [m (ft)]** | | | | | |
| | 2.0 (0.078) | 150 (500) | | TBD | B | 0 | M |
| | 2.0 (0.078) | 450 (1500) | | TBD | B | 0 | N |
| | 3.0 (0.122) | 60 (200) | | TBD | B | 0 | O |
| | 3.0 (0.122) | 300 (1000) | | TBD | B | 0 | P |
| | 4.0 (0.151) | 60 (200) | | TBD | B | 0 | Q |
| | 4.0 (0.151) | 225 (750) | | TBD | B | 0 | R |
| | 5.0 (0.188) | 30 (100) | | TBD | B | 0 | S |
| | 5.0 (0.188) | 150 (500) | | TBD | B | 0 | T |
| | 6.0 (0.220) | 38 (125) | | TBD | B | 0 | U |
| | 6.0 (0.220) | 105 (350) | | TBD | B | 0 | V |
| | 7.0 (0.285) | 45 (150) | | TBD | B | 0 | W |
| | 7.0 (0.285) | 75 (250) | | TBD | B | 0 | X |
| Type III | Custom cut per user's requirements | | | TBD | C | 0 | Y |



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7.2 Source of document.

7.2.1 The Federal Acquisition Regulation may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at http://www.arnet.gov/far.

7.3 National stock numbers (NSNs). See 7.1.

7.4 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.

b. Product conformance provisions (see 3.3).

c. Packaging requirements.

d. Quantity required.

7.5 Key words.
Electrical bonding
Electromagnetic shielding
Gasket material
Monel® matrix
Silicone gel

7.6 Standard sizes. The nominal sizes that are currently available are shown in 7.1.

MILITARY INTERESTS

Custodians:
Navy – SH
Air Force – 99

Review Activity:
DLA – CC

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Preparing Activity:
Navy – SH
(Project 2090-0117-000)



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